Turn in 6

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Problem 1

Since $\sqrt[5]{7}, \zeta\sqrt[5]{7} \in E$, we have

$$\frac{\zeta\sqrt[5]{7}}{\sqrt[5]{7}} = \zeta \in E.$$

Therefore, $\mathbb{Q} \subseteq \mathbb{Q}(\zeta) \subseteq E$ is a chain of field extensios, and thus

$$[E : \mathbb{Q}(\zeta)]\,[\mathbb{Q}(\zeta) : \mathbb{Q}]\,.$$

It suffices to compute these two smaller computations. Note that $\mathbb{Q}(\zeta)$ is generated by the set $Z \coloneqq \left\{\zeta^i \mid 0 \leq i \leq 4, i \in \mathbb{N}\right\}$. Note also that ζ manifestly satisfies the equation $\zeta^5 - 1 = 0$, and that

$$\begin{aligned}(\zeta - 1)\left(\zeta^4 + \zeta^3 + \zeta^2 + \zeta^1 + 1\right) &= \zeta^5 + \zeta^4 + \zeta^3 + \zeta^2 + \zeta - \zeta^4 - \zeta^3 - \zeta^2 - \zeta - 1 \\ &= \zeta^5 - 1 \\ &= 0.\end{aligned}$$

Since ζ is a primitive fifth root of unity, $\zeta \neq 1$, and thus $\zeta - 1 \neq 0$. Dividing by this, we find that

$$\zeta^4 + \zeta^3 + \zeta^2 + \zeta + 1 = 0.$$

Therefore, this set is not linearly independent. Moreover, since $\zeta^5 = 1$, we have

$$\left(\zeta^2\right)^4 + \left(\zeta^2\right)^3 + \left(\zeta^2\right)^2 + \left(\zeta^2\right) + 1 = \zeta^8 + \zeta^6 + \zeta^4 + \zeta^2 + 1 = \zeta^3 + \zeta + \zeta^4 + \zeta^2 + 1 = 0,$$

$$\left(\zeta^3\right)^4 + \left(\zeta^3\right)^3 + \left(\zeta^3\right)^2 + \left(\zeta^3\right) + 1 = \zeta^{12} + \zeta^9 + \zeta^6 + \zeta^3 + 1 = \zeta^2 + \zeta^4 + \zeta + \zeta^3 + 1 = 0,$$

$$\left(\zeta^4\right)^4 + \left(\zeta^4\right)^3 + \left(\zeta^4\right)^2 + \left(\zeta^4\right) + 1 = \zeta^{16} + \zeta^{12} + \zeta^8 + \zeta^4 + 1 = \zeta + \zeta^2 + \zeta^3 + \zeta^4 + 1 = 0.$$

Therefore, $\left\{\zeta, \zeta^2, \zeta^3, \zeta^4\right\}$ form a set of four distinct roots of this polynomial. Since a polynomial in $\mathbb{Q}$ has at most 4 roots, these must be all roots of the polynomial. Since none of these roots are in $\mathbb{Q}$, we find that this polynomial is irreducible over $\mathbb{Q}$, and thus is the minimal polynomial of ζ. Since the polynomial has degree 4, $[\mathbb{Q}(\zeta) : \mathbb{Q}] = 4$.

Now consider E. Note that $x^5 - 7$ is clearly the minimal polynomial of $\sqrt[5]{7}$ over $\mathbb{Q}$. However, note that $x^5 - 7$ is also clearly irreducible over $\mathbb{Q}(\zeta)$ as well, and thus is the minimal polynomial over $\mathbb{Q}(\zeta)$. Since E is the minimal field containing ζ and $\sqrt[5]{7}$, we find that $[E : \mathbb{Q}(\zeta)] = \deg(x^5 - 7) = 5$. Therefore,

$$[E : \mathbb{Q}] = 5 \cdot 4 = 20.$$

Problem 2

Let $K_6 = \mathbb{Q}(\zeta)$. The proof follows from our proof of the above problem.

Let $K_i = \mathbb{Q}(\zeta^i\sqrt[5]{7})$ for $1 \leq i \leq 5$. Since each $\zeta^i\sqrt[5]{7}$ is a root of $x^5 - 7$, which is by Eisenstein's critera irreducible over $\mathbb{Q}$, we find that it is the minimal polynomial for each root over $\mathbb{Q}$, and thus $\left[\mathbb{Q}(\zeta^i\sqrt[5]{7}) : \mathbb{Q}\right] = \deg\left(x^5 - 7\right) = 5$.

Problem 3

Since each K_i for $1 \leq i \leq 5$ does not contain all roots of $x^5 - 7$, the polynomial does not split in K_i. Since this is the minimal polynomial for each of these roots, it follows that no polynomial that doesn't split in $\mathbb{Q}$ can split in these fields. Therefore, these fields are not splitting fields.

The field $\mathbb{Q}(\zeta)$ is a splitting field for $x^4 + x^3 + x^2 + x + 1$, as shown previously. Therefore, it is a splitting field.

Problem 4

Since the degree of E is 20, and since E is clearly Galois as it has characteristic 0 and $x^5 - 7$ splits in it, we have $[E : \mathbb{Q}] = |\mathrm{Gal}(E/\mathbb{Q})| = 20$. Since σ, τ are clearly well-defined as automorphisms by definition, they are elements of $\mathrm{Gal}(E/\mathbb{Q})$. It suffices to show $| \langle \sigma, \tau \rangle | = 20$.

Note that $|\sigma| = 5$ trivially, as σ fixes all elements other than powers of $\sqrt[5]{7}$, and

$$\sigma^5(\sqrt[5]{7}) = \zeta^5 \sqrt[5]{7} = \sqrt[5]{7}.$$

Since this generates all other powers of $\sqrt[5]{7}$, it is easily seen that $|\sigma| = 5$. Similarly, τ fixes $\sqrt[5]{7}$, but it's action on ζ gives it order 4:

$$\tau(\zeta) = \zeta^2, \qquad \tau^2(\zeta) = \left(\zeta^2\right)^2 = \zeta^2,$$

$$\tau^3(\zeta) = \left(\zeta^4\right)^2 = \zeta^8, \qquad \tau^4(\zeta) = \left(\zeta^8\right)^2 = \zeta^{16} = \zeta.$$

Next, note that

$$\left(\tau\sigma\tau^{-1}\right)(\zeta) = \tau\tau^{-1}(\zeta) = \zeta,$$

since σ fixes ζ, and also note that

$$(\tau\sigma\tau^{-1})\left(\sqrt[5]{7}\right) = (\tau\sigma)\left(\sqrt[5]{7}\right) = \zeta^2 \sqrt[5]{7},$$

since τ fixes the radical, and thus so must it's inverse. Also notice that

$$\sigma^2(\zeta) = \zeta, \qquad \sigma^2\left(\sqrt[5]{7}\right) = \zeta^2 \sqrt[5]{7}.$$

Therefore, $\sigma^2 = \tau\sigma\tau^{-1}$. Note that σ^2 generates $\langle\sigma\rangle$, since 2 doesn't divide 5. Therefore,

$$\tau \langle\sigma\rangle \tau^{-1} = \langle\sigma\rangle,$$

and thus we immediately have $\tau^n \langle\sigma\rangle \tau^{-n} = \langle\sigma\rangle$. It follows that $\langle\sigma\rangle$ is normal in $\langle\sigma, \tau\rangle$, and thus

$$\langle\tau\rangle \langle\sigma\rangle = \bigcup_{\tau^i \in \langle\tau\rangle} \tau^i \langle\sigma\rangle = \bigcup_{\tau^i \in \langle\tau\rangle} \langle\sigma\rangle \tau^i = \langle\sigma\rangle \langle\tau\rangle.$$

By a theorem we proved and used extensively last semester, $\langle\sigma\rangle \langle\tau\rangle$ is a subgroup of $\langle\sigma, \tau\rangle$. Also note that since τ fixes the element σ acts on and vice versa, $\langle\tau\rangle \cap \langle\sigma\rangle = 1$ trivially. Therefore, $\langle\tau\rangle \langle\sigma\rangle$ is an internal direct product, and thus

$$\langle\tau\rangle \langle\sigma\rangle \cong \langle\tau\rangle \times \langle\sigma\rangle.$$

This direct product of groups has order $|\langle\tau\rangle| \cdot |\langle\sigma\rangle| = 5 \cdot 4 = 20$, so we have that $\langle\tau, \sigma\rangle = \langle\tau\rangle \langle\sigma\rangle$ since they can both have no more than order 20, since they are subgroups of the Galois group which has order 20. We thus have $|\langle\tau, \sigma\rangle| = 20$, and thus $\langle\tau, \sigma\rangle = \mathrm{Gal}(E/\mathbb{Q})$.